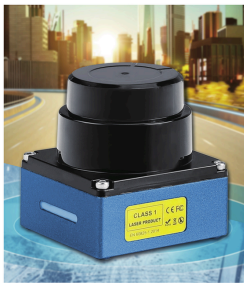


2D 360° SCANNING LIDAR · VF SERIES

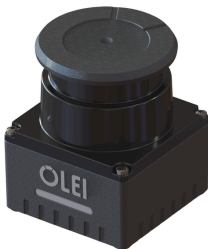
ONE FOOTPRINT OLEI VF SERIES TWO JOBS

Two 360° scanners on one 65 × 65 mm footprint, one sub-5-watt budget, one IP65 rating. The VFD1-10 does the zone logic; the VFH1-50 does the long-range point cloud.

VFD1-10 — 10 m, 16 banks × 3 levels, 4 in / 4 out, ESD Level 4, rain capable. · VFH1-50 — 50 m, 100K pts/s at 0.09°, NTP time sync, UDP / TCP, no switching I/O.



VFD1-10



VFH1-50

■ WHY THIS SENSOR

- **Same footprint, same power budget** — 65 × 65 mm and under 5 W on both — swap one for the other without redesigning the mount
- **VFD1-10: 16 banks × 3 levels, 4 in / 4 out** — NPN or PNP at 50 mA, 130 ms response at 10 Hz — zone decisions without a PLC
- **VFH1-50: 100 000 points per second at 0.09°** — the densest cloud in the OLEI 2D line, with NTP time sync and UDP or TCP output
- **10 m versus 50 m of reach** — and the VFD1-10 holds 10 m at 10% reflectivity, where the VFH1-50 gives 15 m
- **VFD1-10 is ESD hardened and rain capable** — Level 4 per EN IEC 61000-4-2:2009 — 8 kV contact, 15 kV air — with an anti-pollution window
- **VFH1-50 is the lighter, tighter-accuracy option** — < 500 g and < ±20 mm absolute from 2% to 80% reflectivity
- **Both: 360° scanning, IP65, 80 000 lux, -10 to +50 °C** — reflector and natural navigation from a single sensor
- **Both configure from OLEI's own tooling** — SafetyConfigTool and OLAMView on the VFD1-10, OleView on the VFH1-50

360°

BOTH SENSORS

10 m / 50 m

WORKING RANGE

14.4K / 100K

POINTS PER SECOND

65 × 65 mm

SHARED FOOTPRINT

■ APPLICATION SCENARIOS



AMR & AGV

Full-circle obstacle avoidance from one chassis mount



SURVEY & MAPPING

360° natural-feature and reflector mapping at range



HYBRID NAVIGATION

Reflector and natural navigation from one sensor



INTRALOGISTICS

Forklifts, tuggers and conveyor zones in active aisles

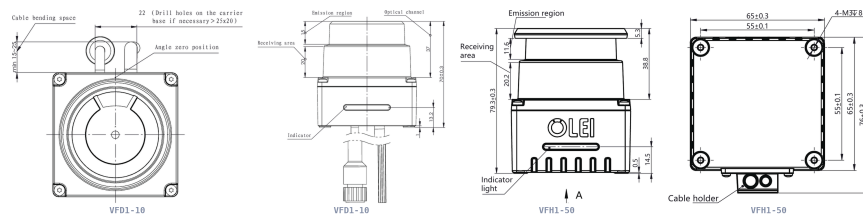
VF SERIES — SPECIFICATIONS



PARAMETER	VFD1-10	VFH1-50
Sensing technology	2D scanning ToF LiDAR	
Laser wavelength	905 ± 20 nm	
Laser class	Class 1 (IEC 60825-1:2014)	Class I
Channels	1	
Field of view (horizontal)	360°	
Beam divergence	8 mrad (H) · 3 mrad (V)	< 5 mrad
Scanning frequency	10 / 15 / 25 Hz	15 / 20 / 25 Hz
Angle resolution	0.25° @ 10 Hz 0.5° @ 15 Hz · 1° @ 25 Hz	0.09°
Working distance (based on reflectivity)	80% / 10% / reflector 0.1–10 m 4% 0.1–5 m	80% / reflector 0.1–50 m 10% 0.1–15 m · 4% 0.1–5 m
Distance resolution	1 mm	
Absolute accuracy ①	< ±30 mm (0.4–10 m)	< ±20 mm (2%–80%) Reflector < ±20 mm
Repeat accuracy ①	≤ 20 mm (0.4–10 m)	≤ 20 mm (0.4–50 m)
Point cloud density	14.4K @ 10 Hz 10.8K @ 15 Hz · 9K @ 25 Hz	100K @ 25 Hz
Horizontal error (horizontal angle)	≤ ±1°	
Response time	130 ms @ 10 Hz	—

PARAMETER	VFD1-10	VFH1-50
Number of bank fields	16 banks, 3 levels/bank	—
Switching inputs / outputs	4 / 4	—
Output mode	NPN / PNP (hardware optional)	—
Output current	50 mA	—
Ethernet	100 Mbps Ethernet	
Output data protocol	—	UDP, TCP
Time synchronisation	—	NTP
Power connection	12-core adapter with cable 20 cm, TPU	DC 5.5 × 2.1 connector Pigtail, 50 cm
Data connection	4-core adapter, M12×1 female 200 cm, TPU	RJ45 network port Pigtail, 50 cm
Operating voltage	12–26 V DC	12–24 V
Power consumption @ 25 °C	< 5 W (no load)	< 5 W
Mechanical dimensions	65 × 65 × 70 mm	< 65 × 65 × 79.5 mm
Weight	300 g ± 5%	< 500 g
Degree of protection	IP65	
Operating temperature	–10 to +50 °C	
Storage temperature	–20 to +70 °C	–20 to +70 °C
Ambient light limit	80 000 lux	
Software	SafetyConfigTool · OLAMView	OleView

OUTLINE AND MOUNTING



UNIT: MM · DRAWINGS NOT TO SCALE

Certification. Both: RoHS, CE, FCC, FDA. EMC per IEC 61000-6-2:2016-08 and IEC 61000-6-3:2006-07. Vibration per IEC 60068-2-6:2007. Shock per IEC 60068-2-27:2008. VFD1-10 additionally: ESD per EN IEC 61000-4-2:2009 Level 4 — 8 kV contact discharge, 15 kV air discharge — and stated detection capability in moderate or light rain.

Functional safety. The manufacturer's specification for this product states laser safety Class 1 per IEC 60825-1:2014 and the EMC, vibration and shock standards listed above. It does not state an IEC 61496 Type rating or an ISO 13849 performance level. Treat the zone outputs as obstacle-avoidance and area-monitoring signals, not as a rated safety function, and do not use this sensor as the sole protective device. Where a rated scanner is required, ask MorpheusTEK about the OLEI GS1-5.

Notes. ① Standard values; actual values depend on environmental conditions, and both sensors specify accuracy from 0.4 m rather than from the 0.1 m minimum range. A dash means the parameter does not apply to that model. On the VFD1-10, angle resolution and point rate both fall as scan frequency rises. The VFD1-10 holds its full 10 m at 10% reflectivity; the VFH1-50 drops to 15 m at 10%, so on dark targets the gap between the two is much smaller than the headline 10 m versus 50 m.

Installation. Both are 360° sensors and need the full circle clear — mount above the platform's own structure rather than recessed into it, or the blind sectors will be created by the vehicle. Allow the stated cable bending space beneath the VFD1-10. Leads are short on both models — 20 and 200 cm on the VFD1-10, 50 cm on the VFH1-50 — so plan a junction close to the mount. Clean the optical window on a schedule in dirty environments.

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